Exploring the Effects of Terrain Attributes

Understanding the Spatial Distribution of Soil Geochemical and Colour Properties

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Acknowledgements

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Introduction

- Quantifying the spatial variability and distribution of soil proprties is important:
    - Agronomic production
    - Agri-environmental research
    - Modelling exercises
- Used to understand field/watershed processes and guide management practices

Characterizing soil properties

- Focus has been on:
  - Agronomically important nutrients (N, P, K, Ca, S)
  - Pedological and weathering processes
  - Soil surveying and mapping
  - Pollution/contamination issues

Research questions

1. How do soil geochemical and colour properties vary across different land uses?

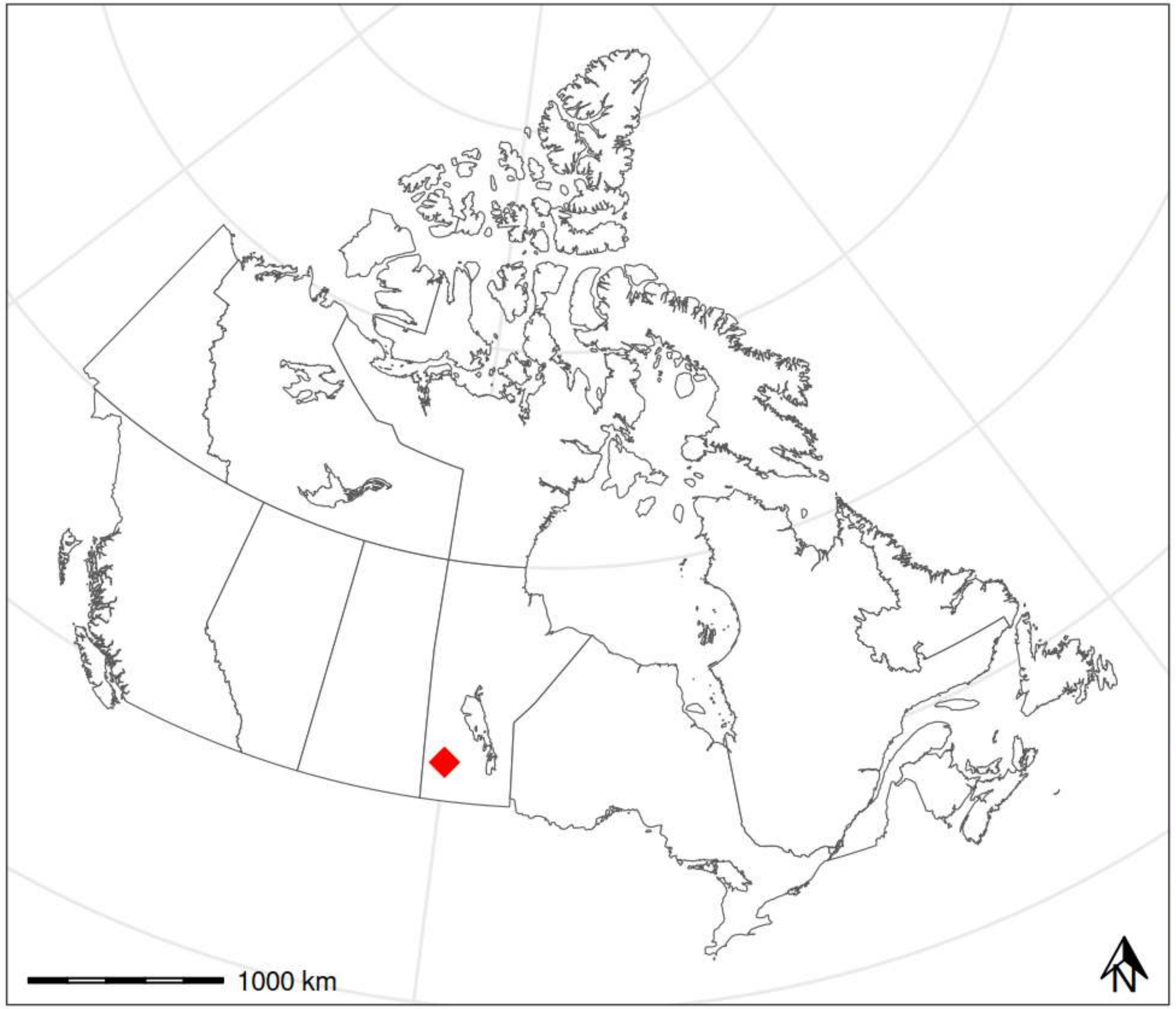
2. Which terrain attributes best explain the spatial patterns of these soil properties?

Objectives

Using a range of soil colour and geochemical properties across two contrasting land uses:

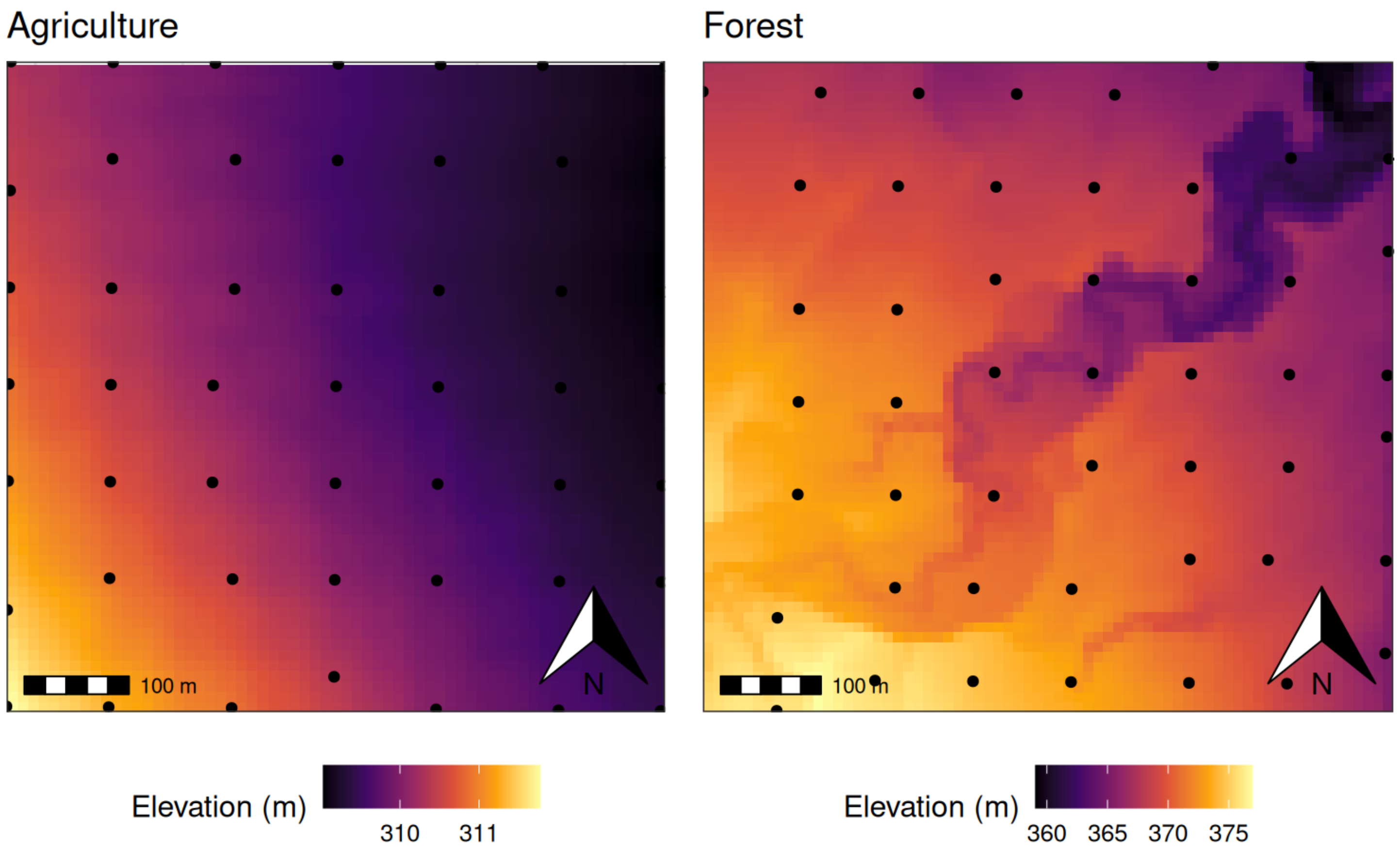
1. Quantify the variability
2. Characterize the spatial patterns
3. Assess the the importance of terrain attributes

Location



Sampling

- Surface soil
- 49 points at 100m spacing



Lab analysis

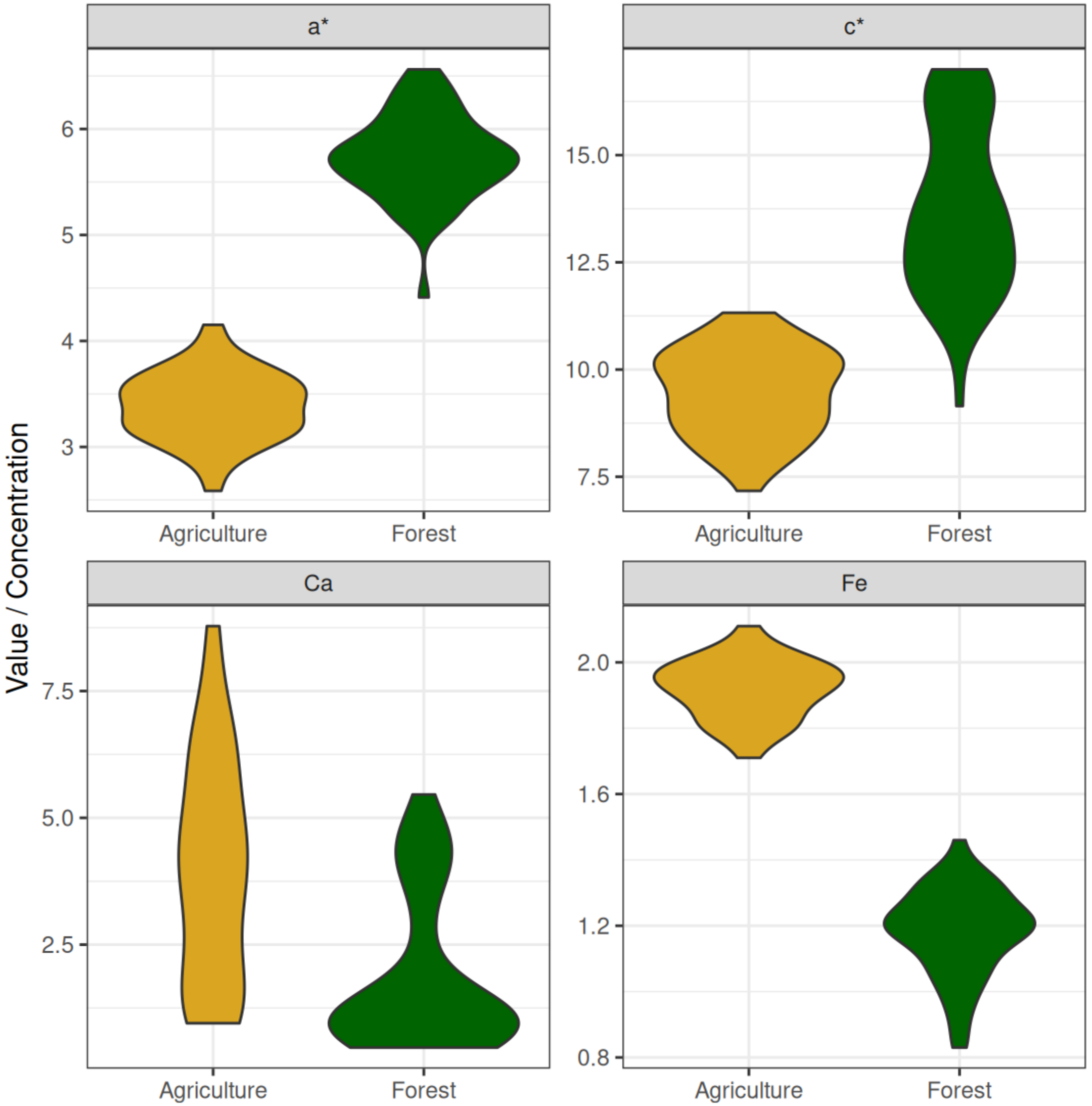
- Sieved to < 63 um
- Geochemistry
  - Aqua-regia
  - 51 geochemical elements
- Spectral reflectance
  - FieldSpecPro
  - 15 colour coefficients

Based on previous work (Luna Miño et al. 2024)

- Ca, Co, Cs, Fe, Li, La, Nb, Ni, Rb, and Sr
- *a*, b*, h*, and x*

Univariate analysis

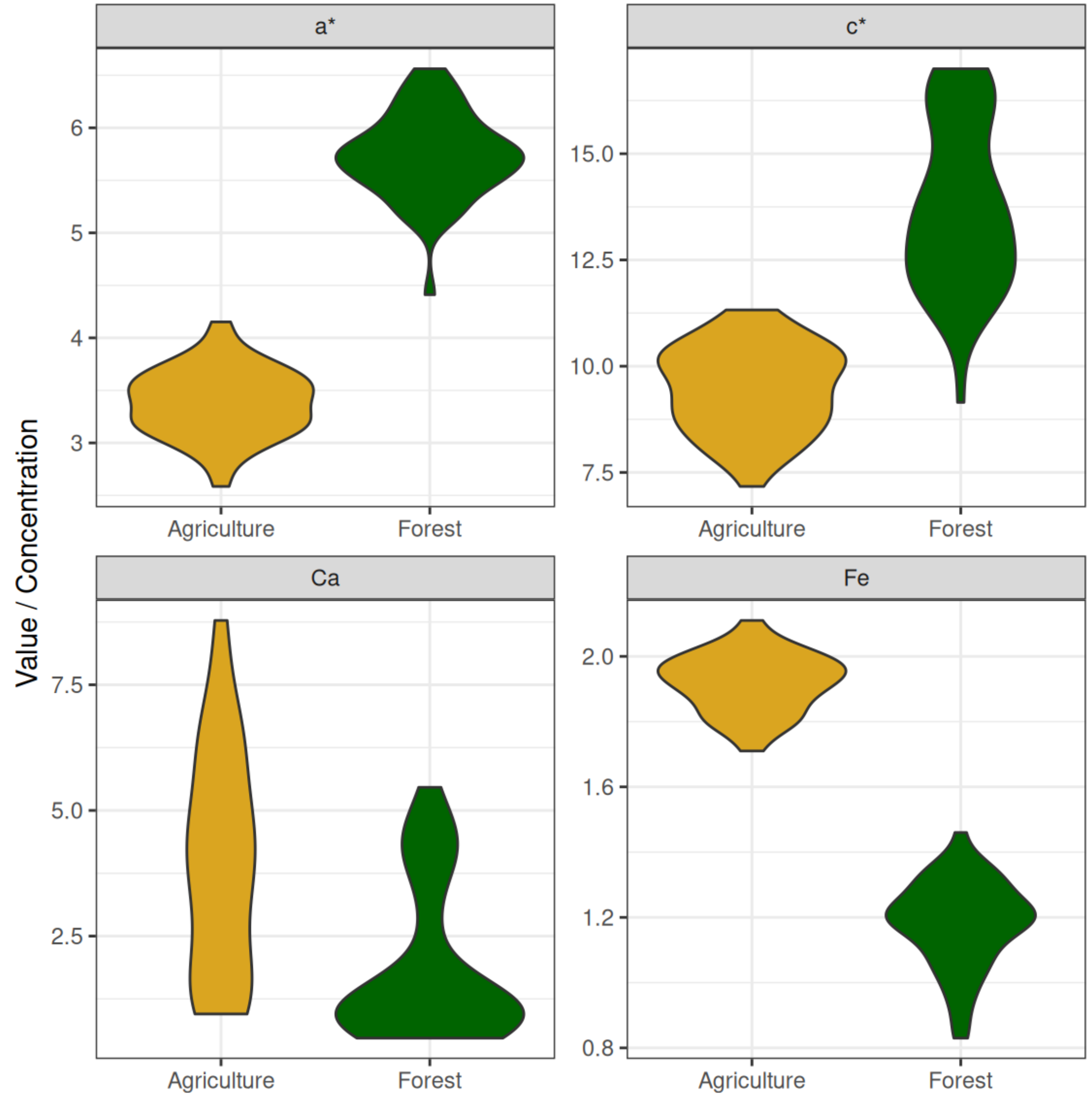
- Mean
- Standard deviation
- Skewness
- Coefficient of variation



Univariate analysis

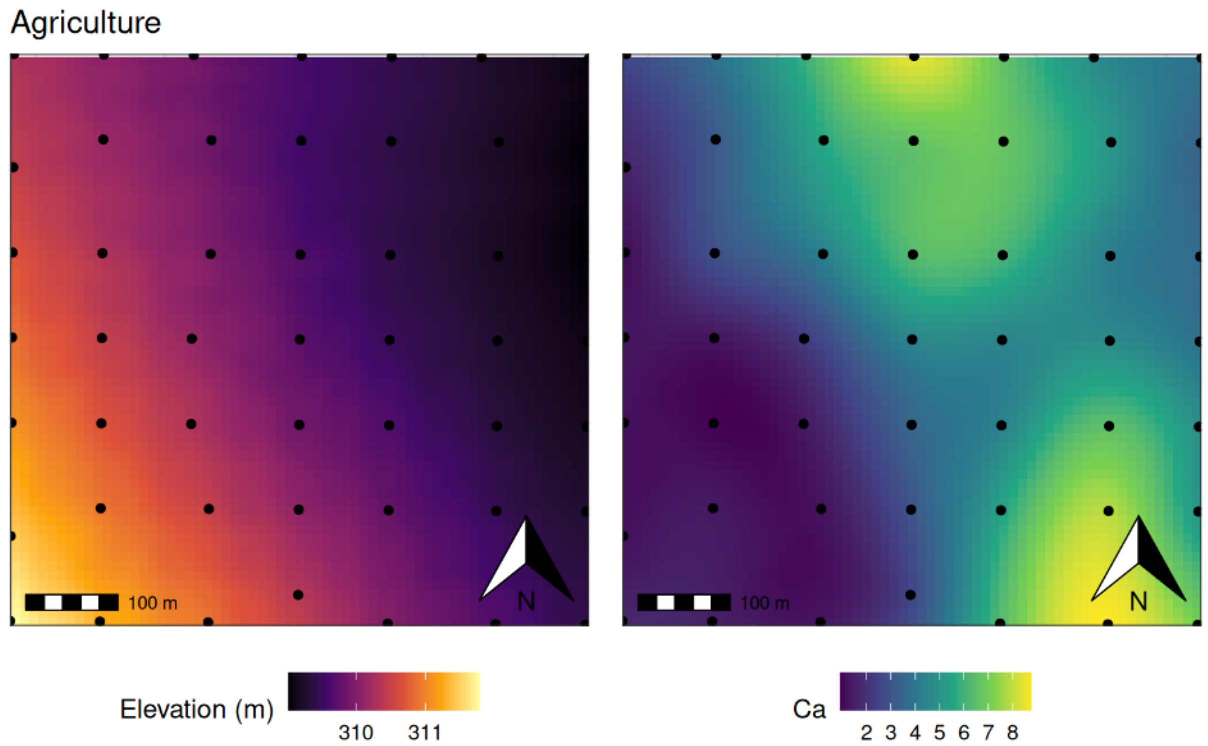
Overall

- **Colour** properties and the **agricultural** land use
  - Exhibited lower variability and more symmetrical data
- Forested site has a more complex topography and geomorphic setting (floodplain)
  - Greater variability in SOM and grain size



Geospatial analysis

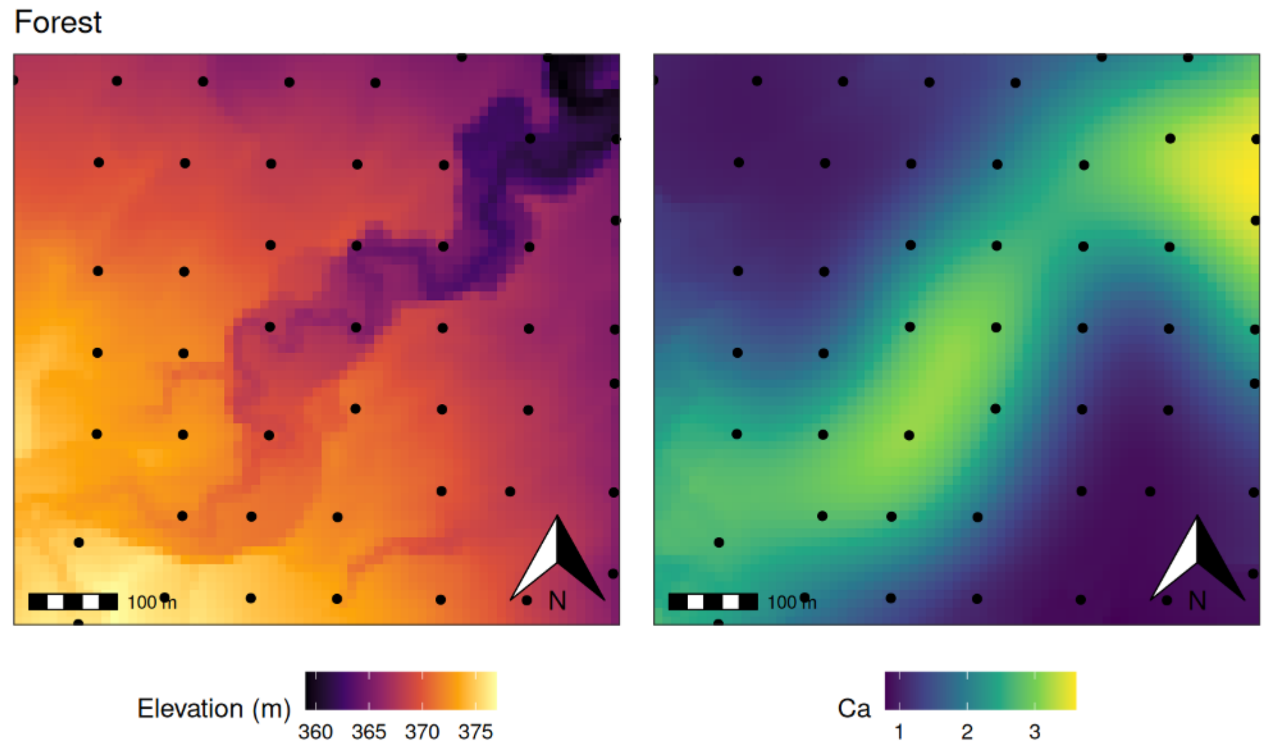
- Spatial autocorrelation
  - Semivariograms
- Interpolation and mapping
  - Kriging



Nugget = 0.0
Sill = 7.2
Range = 580m
Spatial Class = Strong

Geospatial analysis

- Spatial autocorrelation
  - Semivariograms
- Interpolation and mapping
  - Kriging

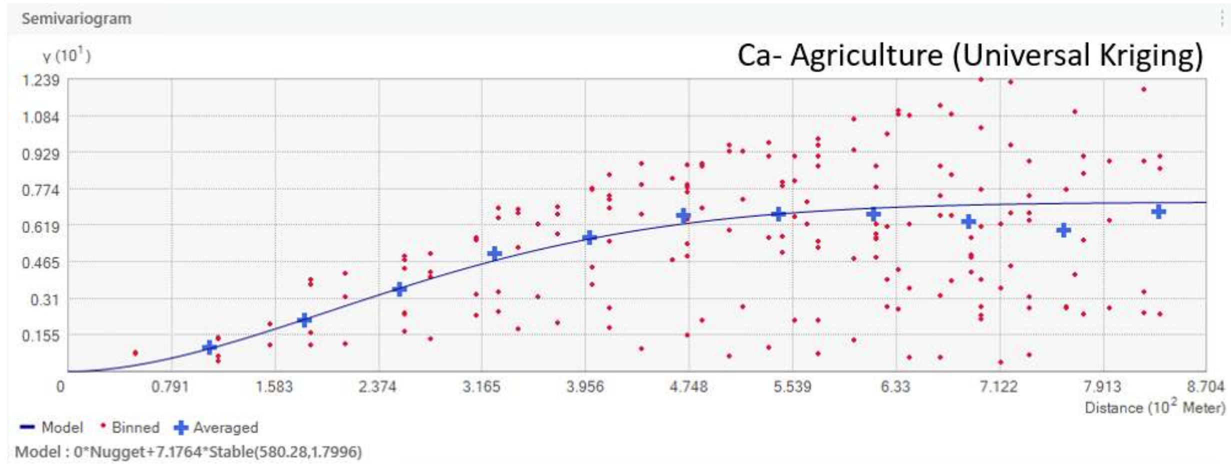


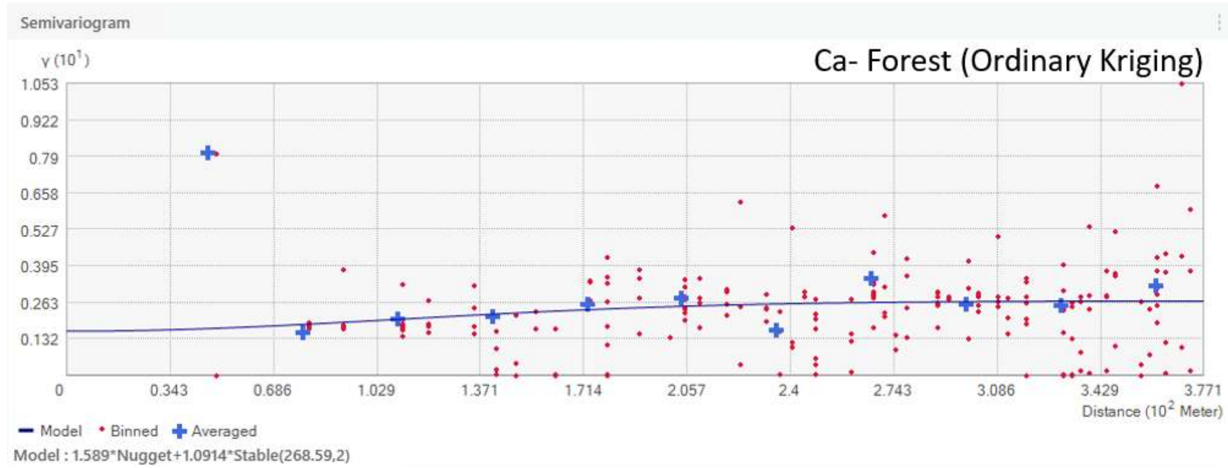
Nugget = 1.6
Sill = 2.7
Range = 269m
Spatial Class = Moderate

Geospatial analysis

Semivariogram interpretation

- Small nugget reflects low measurement or sampling error
- Small sill indicates low overall variance
- Small range indicate spatial correlation persists over short distances

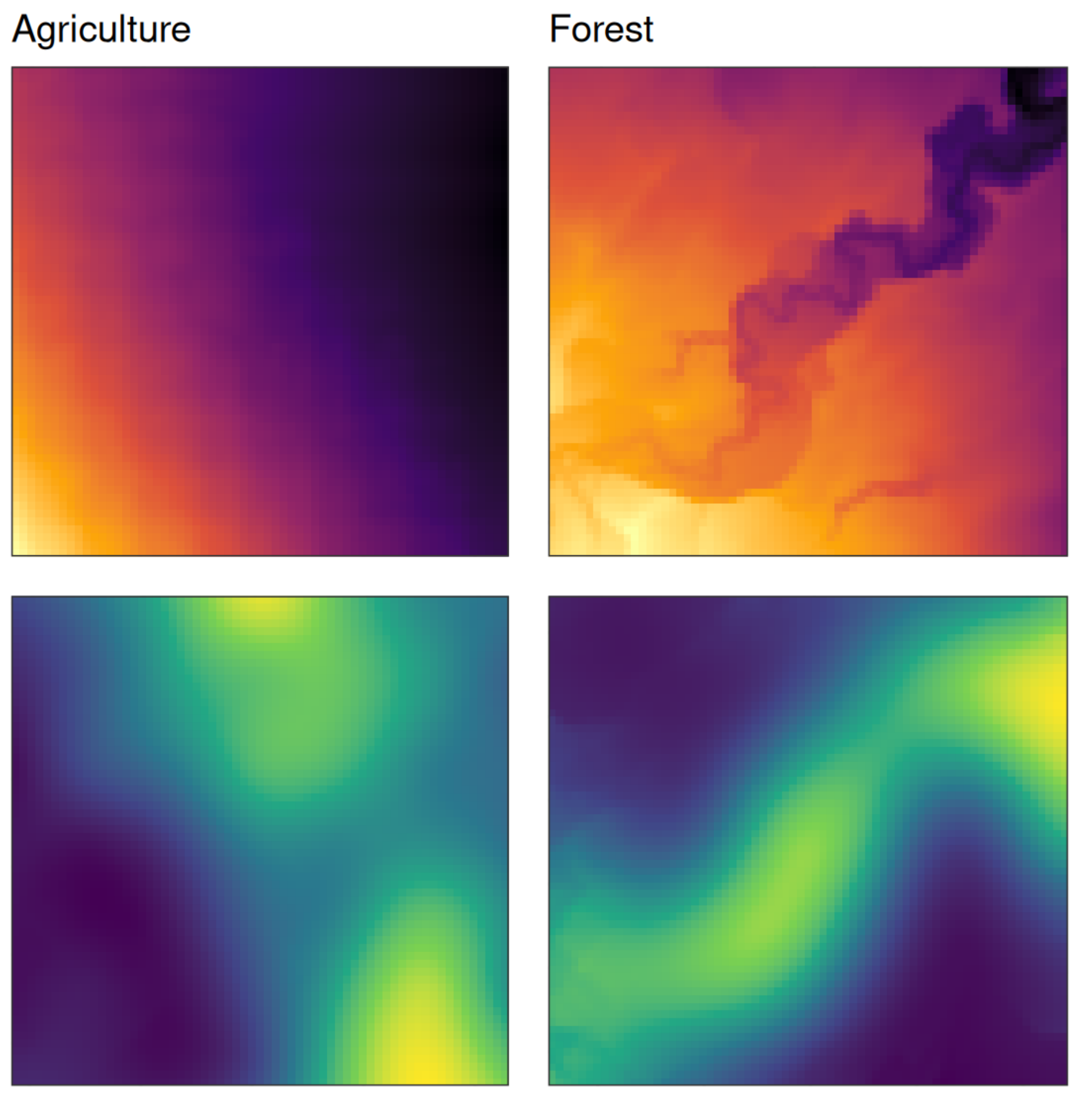




Geospatial analysis

Spatial autocorrelation

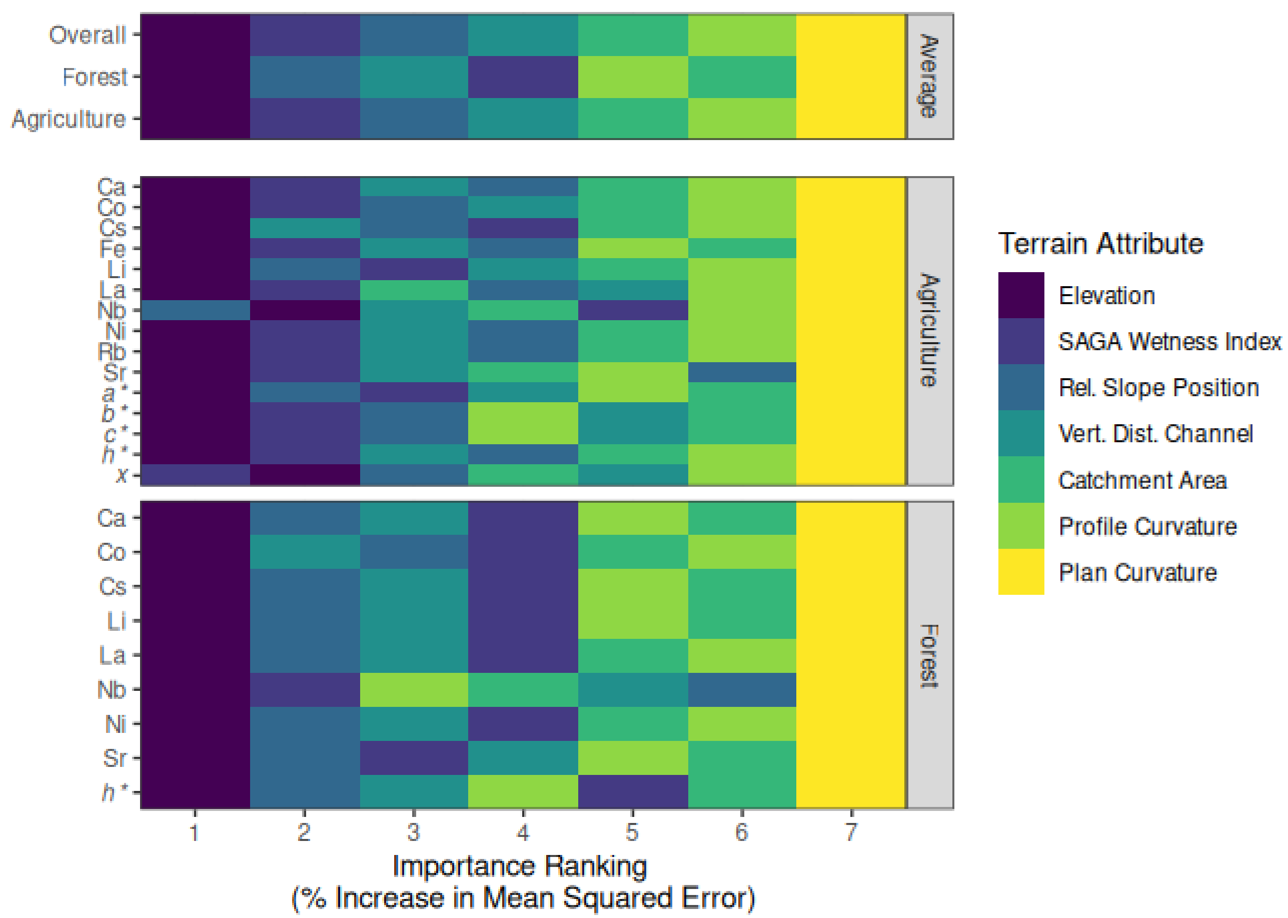
- Soil properties at the **agricultural site** exhibited **stronger** spatial autocorrelation
  - 6 soil properties at the forested site exhibited **no** spatial autocorrelation
- Soil properties presented patterns that roughly matches the topography



Terrain analysis

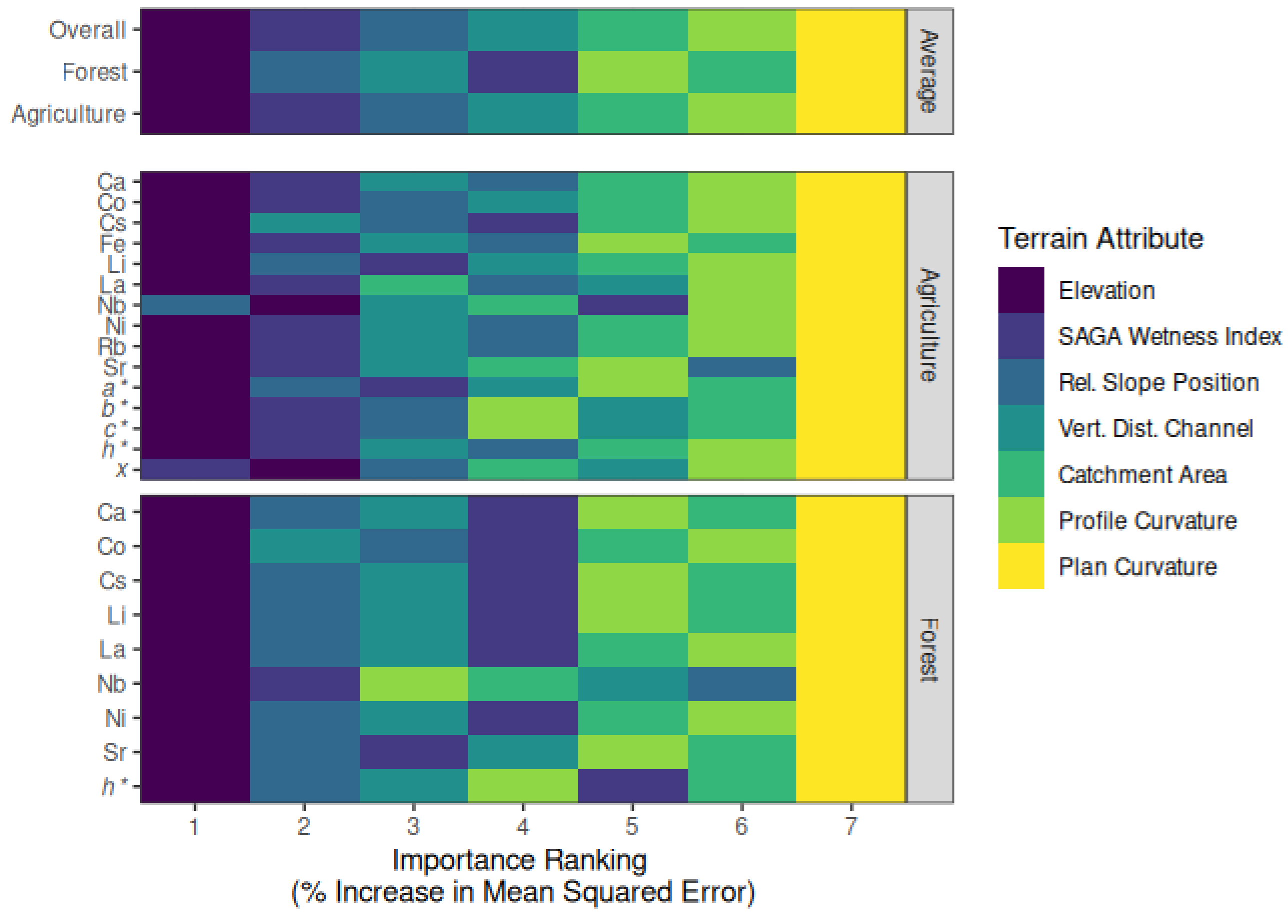
- Terrain attributes
  - System for Automated Geoscientific Analyses (SAGA)
  - Random Forest Regression

1. Plan curvature
2. Profile curvature
3. SAGA wetness index
4. Catchment area
5. Relative slope position
6. Vertical channel network distance



Terrain analysis

- Elevation was ranked as the most important predictor
  - SAGA Wetness Index
  - Relative Slope Position
- Patterns linked to hydrologic properties and processes
- Terrain attributes can be used to guide sampling and interpret data



Conclusions

- Agricultural site:
  - Gently sloping terrain
  - Lower variability
  - Approximately normal data distributions
  - Moderate to strong spatial autocorrelation

- Forested site:
  - Complex terrain
  - Higher variability
  - Data often non-normal
  - Fewer properties with spatial autocorrelation

Conclusions

- Topographic effects evident in many soil property patterns
- Top terrain predictors: elevation, SAGA Wetness Index, and relative slope position
  - Terrain–soil relationships were inconsistent in strength and direction
- Terrain-driven spatial patterns can inform more targeted soil sampling

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Want to learn more?

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